



US0D1091991S

(12) **United States Design Patent** (10) **Patent No.: US D1,091,991 S**
Williams et al. (45) **Date of Patent: ** Sep. 2, 2025**

(54) **VACUUM CONDUIT CONNECTOR** 3,955,237 A 5/1976 Chateaufeuf et al.
4,053,962 A 10/1977 McDowell
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(US) 4,262,542 A * 4/1981 Freund, Jr. G01F 1/58
73/861.12
D264,011 S 4/1982 Genoa
4,557,535 A * 12/1985 Keane A47L 9/246
439/13
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(**) Term: **15 Years**
(21) Appl. No.: **29/966,689**
(22) Filed: **Oct. 4, 2024**
6,478,342 B1 11/2002 Berfield
6,581,974 B1 6/2003 Ragner et al.
6,648,252 B2 * 11/2003 Strutz E03C 1/2665
241/101.2
6,785,934 B2 9/2004 Bruno et al.
6,820,305 B2 11/2004 Albert et al.
6,836,929 B2 1/2005 Kim
D534,760 S * 1/2007 Patterson D7/375
D534,761 S * 1/2007 Patterson D7/375
7,159,270 B2 1/2007 Genoa et al.
7,231,686 B1 6/2007 Matheney
D610,760 S * 2/2010 Zugen D32/31
7,934,286 B2 5/2011 Yoo et al.
8,167,337 B2 5/2012 Bruno et al.
8,555,461 B2 10/2013 Hachtmann
8,789,854 B2 7/2014 Christian, Jr. et al.
D729,926 S * 5/2015 Tomasiak D23/388
D792,637 S 7/2017 Godlewski et al.
9,731,053 B2 8/2017 Alai
D803,493 S 11/2017 Lutz et al.
D805,612 S 12/2017 Mursu et al.
D809,723 S 2/2018 Pierzynski et al.
D812,331 S 3/2018 LaBarbera et al.
10,080,471 B2 9/2018 Reimer et al.
10,238,256 B2 3/2019 Cole et al.
10,278,555 B2 5/2019 Rupp et al.
D851,343 S 6/2019 Hinchliffe
10,443,768 B1 10/2019 McConnell et al.
10,702,111 B2 7/2020 Tomasiak et al.
10,743,734 B2 8/2020 Chavana, Jr. et al.

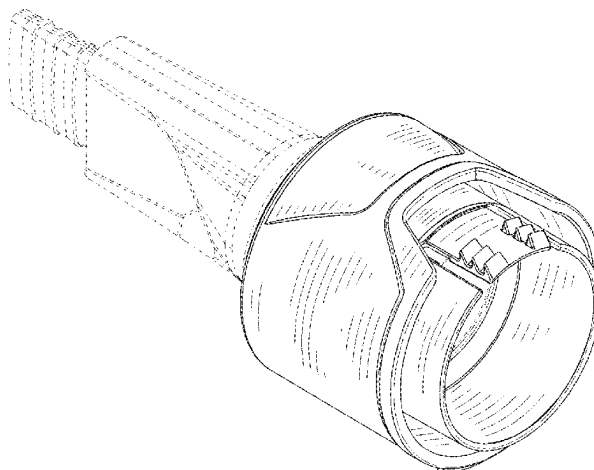
Related U.S. Application Data

(62) Division of application No. 29/873,450, filed on Mar.
31, 2023, now Pat. No. Des. 1,051,534.
(51) **LOC (15) Cl.** **15-05**
(52) **U.S. Cl.**
USPC **D32/32**
(58) **Field of Classification Search**
USPC D8/382, 394, 396, 499; D32/25, 31, 32,
D32/39
CPC . A47L 9/00; A47L 9/02; A47L 9/0639; A47L
9/24; A47L 9/242; A47L 9/248; A47L
11/4094
See application file for complete search history.

References Cited

U.S. PATENT DOCUMENTS

2,605,163 A 7/1952 Lee
2,692,781 A * 10/1954 Langille A47L 9/1427
285/252
3,127,227 A 3/1964 Edwards



10,869,586	B2	12/2020	Bearup et al.	
D910,255	S *	2/2021	Bloemendaal	D32/31
D914,309	S	3/2021	Cole et al.	
11,160,426	B1	11/2021	Williamson	
11,214,974	B2	1/2022	Smith	
11,253,125	B1	2/2022	Hoyt	
11,399,683	B2	8/2022	Cleff	
2004/0025286	A1	2/2004	Boys	
2005/0005389	A1	1/2005	Rau et al.	
2008/0216280	A1	9/2008	Marshall et al.	
2012/0104744	A1	5/2012	Petty	
2015/0198274	A1	7/2015	Farland et al.	
2019/0223671	A1 *	7/2019	Tomasiak	A47L 9/02
2019/0387936	A1	12/2019	Richey et al.	
2021/0235952	A1	8/2021	Nieschwitz et al.	
2022/0071462	A1	3/2022	Hillard	
2022/0167812	A1	6/2022	Droguett Larrain	
2022/0183520	A1	6/2022	Reeder et al.	
2022/0322899	A1	10/2022	Reeder et al.	

FOREIGN PATENT DOCUMENTS

CN	201444537	U	4/2010	
CN	101427895	B	1/2012	
CN	202669468	U	1/2013	
CN	106805842	A	6/2017	
CN	207748083	U *	8/2018	
CN	207777183	U *	8/2018	
CN	208335757	U	1/2019	
CN	216724408	U *	6/2022	
CN	217176830	U	8/2022	
CN	217898374	U *	11/2022	
CN	218325180	U	1/2023	
DE	20215373	U1	12/2002	
EP	0181694	A1	5/1986	
EP	666051	A1 *	8/1995	A47L 9/242
EP	1062902	A2	12/2000	
EP	1088624	A2	4/2001	
EP	1321088	A2	6/2003	
GB	2119046	A *	11/1983	A47L 9/242
IN	188211	B *	8/2002	
JP	2003049771	A *	2/2003	
JP	2003093293	A *	4/2003	
JP	2009170232	A *	7/2009	
KR	20010011665	A	2/2001	
KR	467383	Y1 *	6/2013	
KR	2015006137	A *	1/2015	
KR	2015006137	A	1/2015	
KR	20170001718	U	5/2017	
WO	2020264183	A1	12/2020	
WO	2020264205	A1	12/2020	

OTHER PUBLICATIONS

Adjustable Connector, posted Oct. 24, 2019 [online], [retrieved Jan. 15, 2025]. Retrieved from internet, <https://www.amazon.com/Fielect-Waterproof-Corrugated-Adjustable-Locknut/dp/B07ZHYYGHC> (Year: 2019).*

Tube Connector, posted Mar. 22, 2019 [online], [retrieved Jan. 15, 2025]. Retrieved from internet, <https://www.amazon.com/uxcell-Corrugated-Connector-AD34-5-Joints/dp/B07PYWR34L> (Year: 2019).*

Connecting Sleeve, posted Dec. 18, 2017 [online], [retrieved Jun. 11, 2024]. Retrieved from internet, <https://www.amazon.com/Festool-202346-Reducing-Sleeve-DM-AS/dp/B07924Q5T2> (Year: 2017).

"1 $\frac{1}{8}$ in. Dust Extraction Adapter for Wet/Dry Shop Vacuums (3-Piece)", Home Depot product webpage, 5 pages, retrieved online on Mar. 30, 2023 at URL: <https://www.homedepot.com/p/Milwaukee-1-7-8-in-Dust-Extraction-Adapter-for-Wet-Dry-Shop-Vacuums-3-Piece-49-90-1980/319226245>.

"3 Pieces Vacuum Hose Adapter, 2 $\frac{1}{2}$ Inch to 1 $\frac{1}{4}$ Inch, 1 $\frac{1}{4}$ Inch to 1 $\frac{3}{8}$ Inch to 1 $\frac{1}{2}$ Inch, 1 $\frac{3}{8}$ Inch to 1 $\frac{1}{4}$ Inch Wet Dry Vacuum Converter Reducer Attachments for Vacuum Cleaner", Amazon product webpage, 10 pages, retrieved online on Mar. 30, 2023 at

URL: https://www.amazon.com/Universal-Adapter-Converter-Reducer-Attachments/dp/B08Y8PBKHW/ref=sr_1_8keywords=2+1%2F2+inch+to+1+7%2F8+inch+vacuum+hose+adapter+qid=1665389458prefix=1-7%2F8+2-1%2F2+inch+vacuum+%2Caps%2C346+sr=8-8.

"Air-Tip 1 $\frac{1}{4}$ in.-2 $\frac{1}{2}$ in. Conduit Line Puller Attachment For Wet/Dry Shop Vacuums (3-Piece)", Home Depot product webpage, 6 pages, retrieved online on Mar. 30, 2023 at URL: <https://www.homedepot.com/p/Milwaukee-AIR-Tip-1-1-4-in-2-1-2-in-Conduit-Line-Puller-Attachment-For-Wet-Dry-Shop-Vacuums-3-Piece-49-90-2024/319238054>.

"Hose and Accessory Adapter Kit for Ridgid Wet/Dry Shop", Home Depot product webpage, 9 pages, retrieved online on Mar. 30, 2023 at URL: <https://www.homedepot.com/p/RIDGID-Hose-and-Accessory-Adapter-Kit-for-RIDGID-Wet-Dry-Shop-Vacuums-VT1755/202077239>.

"Hose and Accessory Adapter Kit for Wet/Dry Shop Vacuums (3-Piece)", Home Depot product webpage, 5 pages, retrieved online on Mar. 30, 2023 at <https://www.homedepot.com/p/Milwaukee-Hose-and-Accessory-Adapter-Kit-for-Wet-Dry-Shop-Vacuums-3-Piece-49-90-1991/319229172#overlay>.

* cited by examiner

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Assistant Examiner — Taylor O Mortorff

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(57)

CLAIM

The ornamental design for a vacuum conduit connector as shown and described.

DESCRIPTION

FIG. 1 is a perspective view of a first embodiment of a vacuum conduit connector showing the new design.

FIG. 2 is a front view thereof.

FIG. 3 is rear view thereof.

FIG. 4 is a right-side view thereof.

FIG. 5 is a left-side view thereof.

FIG. 6 is a top view thereof.

FIG. 7 is a bottom view thereof.

FIG. 8 is a sectional view thereof, taken along line 8-8 in FIG. 6.

FIG. 9 is a perspective view of a second embodiment of a vacuum conduit connector showing the new design.

FIG. 10 is a front view thereof.

FIG. 11 is rear view thereof.

FIG. 12 is a right-side view thereof.

FIG. 13 is a left-side view thereof.

FIG. 14 is a top view thereof.

FIG. 15 is a bottom view thereof; and,

FIG. 16 is a sectional view thereof, taken along line 16-16 in FIG. 14.

In these drawings, the solid lines illustrate the claimed ornamental design, whereas broken lines, if present, illustrate unclaimed subject matter and form no part of the claimed design. Stippling in the drawings is used to indicate a contrast in appearance, with no specific color, material, texture, or other feature being claimed. The dash-dot lines shown in FIGS. 6 and 14 illustrate the location from which the section drawings shown in FIGS. 8 and 16 are taken. The dash-dot lines form no part of the claimed design.

1 Claim, 16 Drawing Sheets

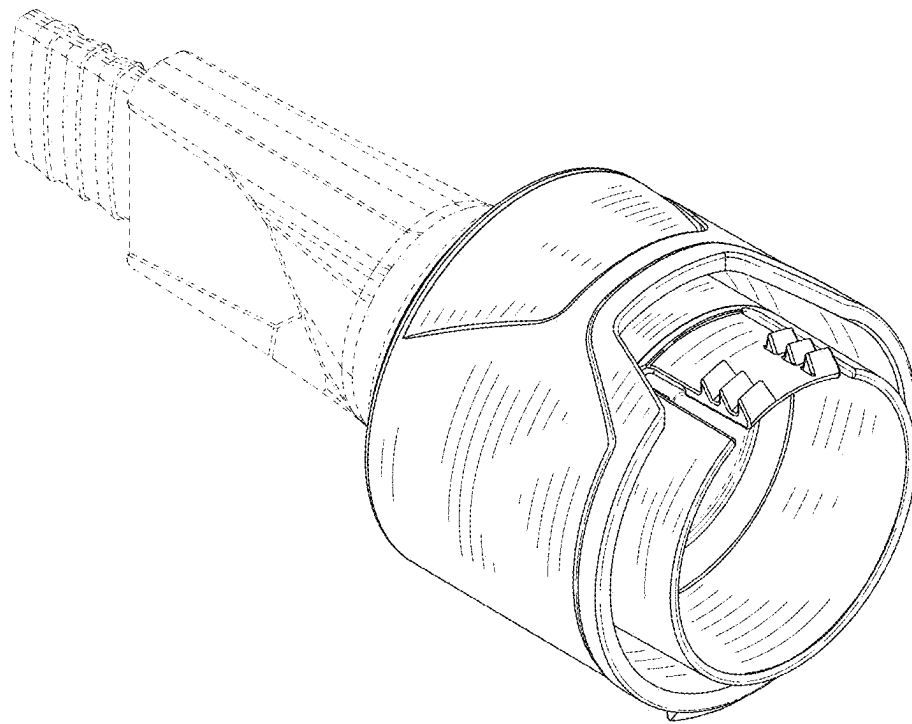


FIG. 1

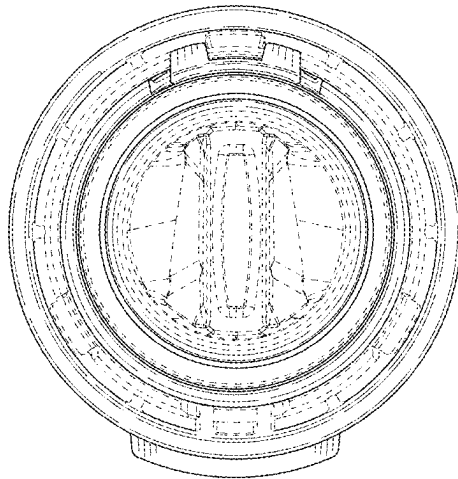


FIG. 2

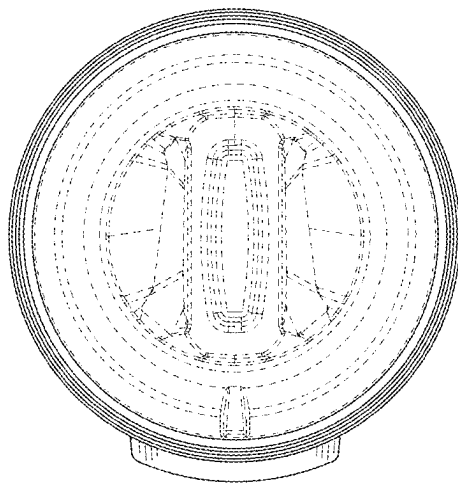


FIG. 3

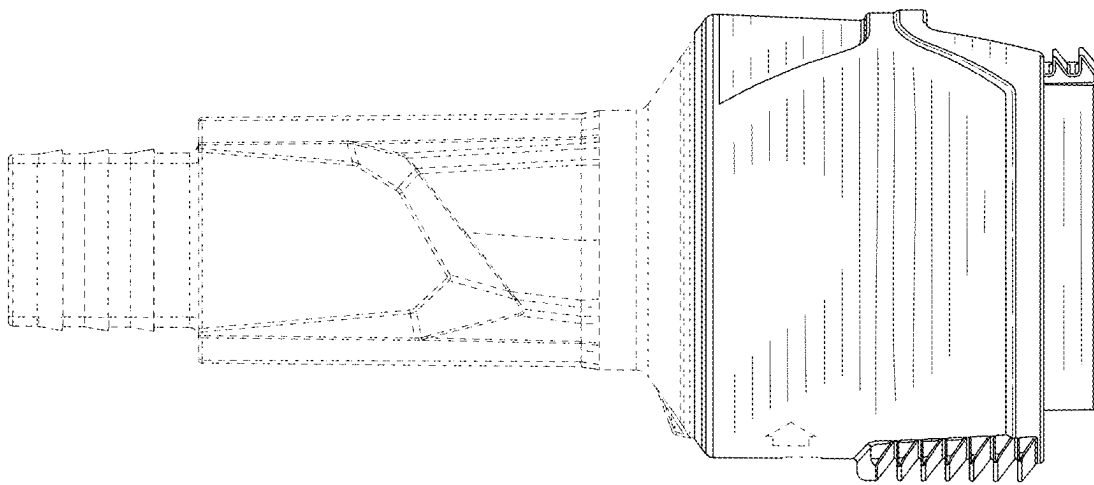


FIG. 4

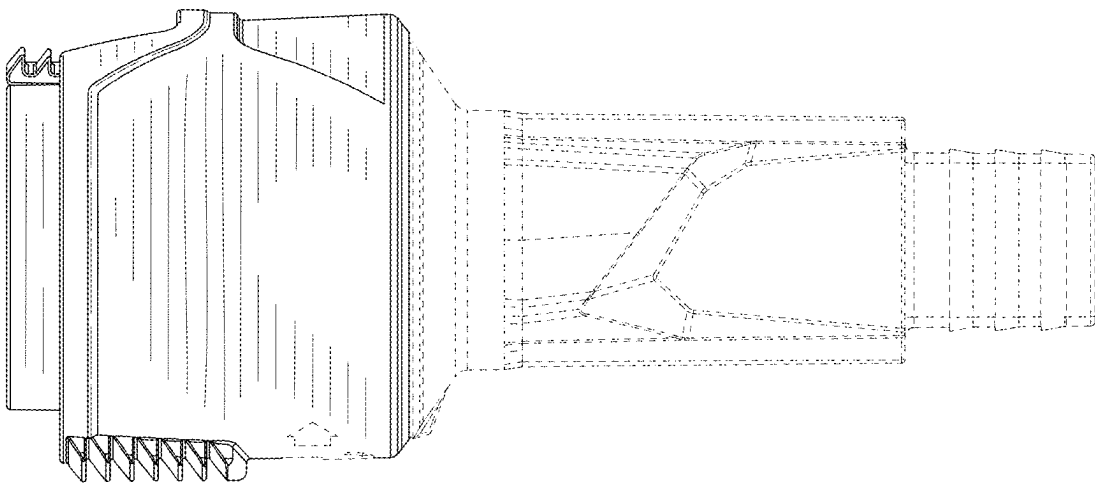


FIG. 5

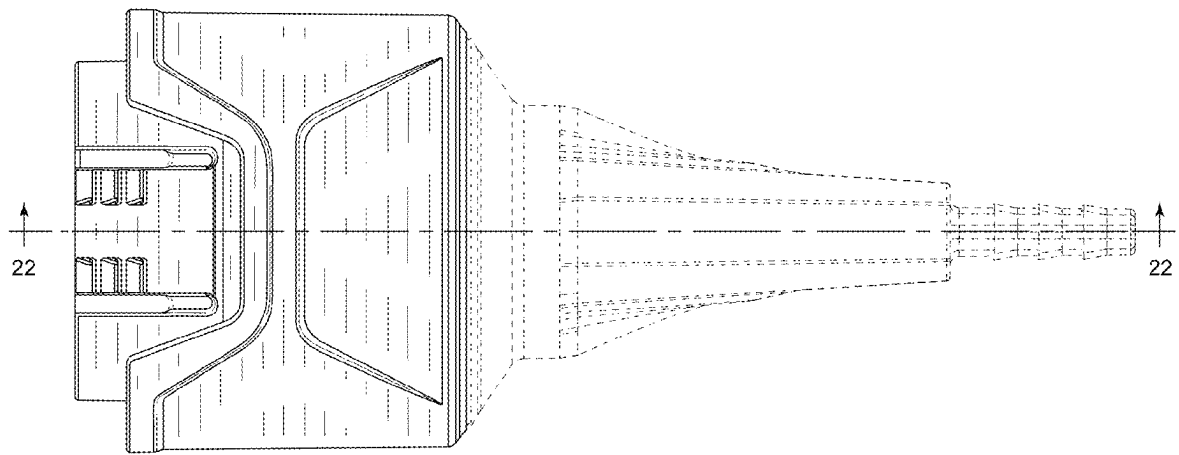


FIG. 6

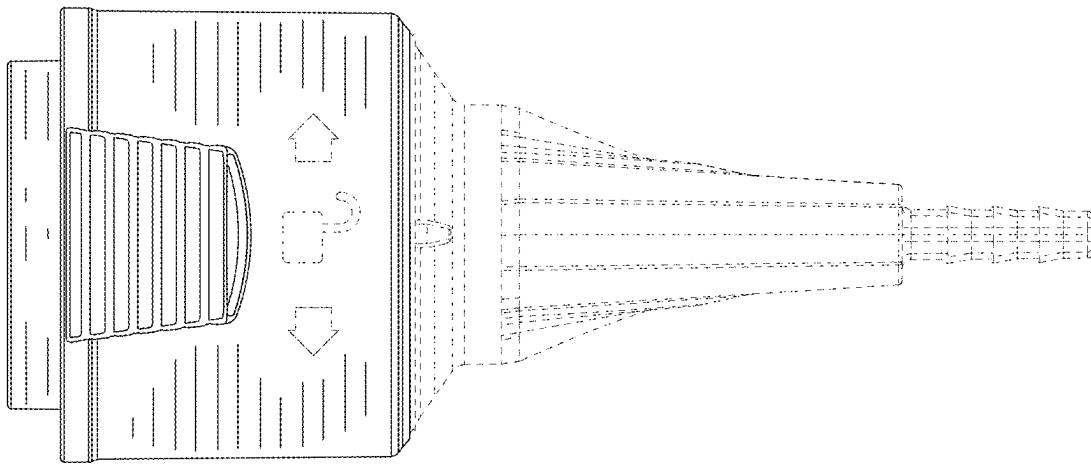


FIG. 7

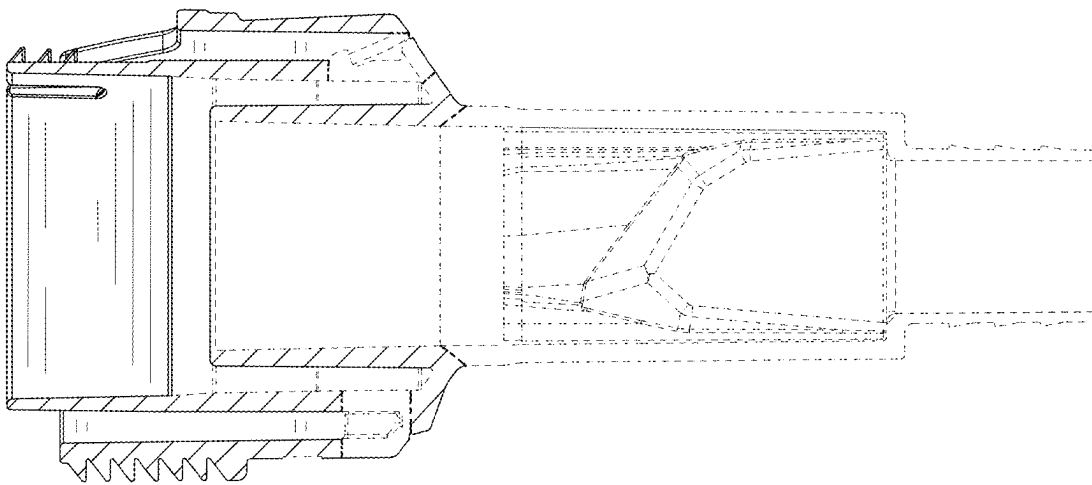


FIG. 8

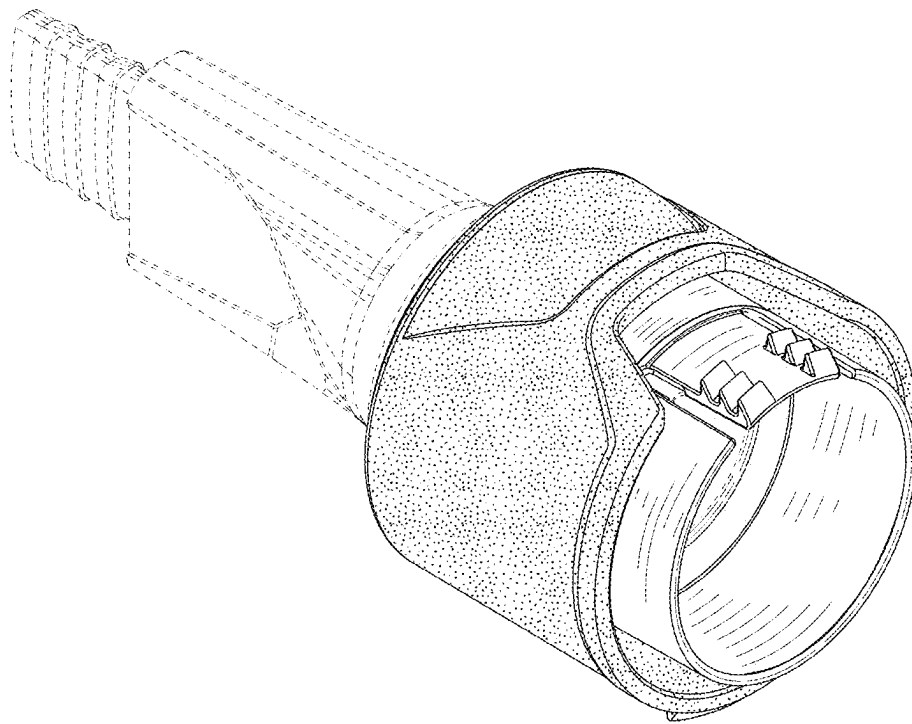


FIG. 9

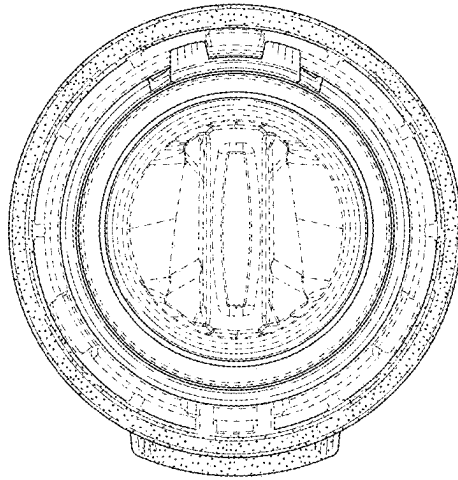


FIG. 10

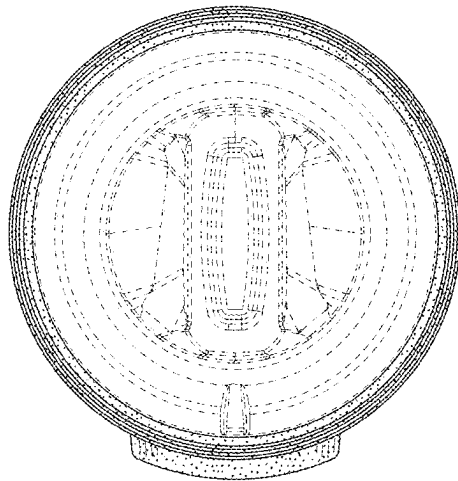


FIG. 11

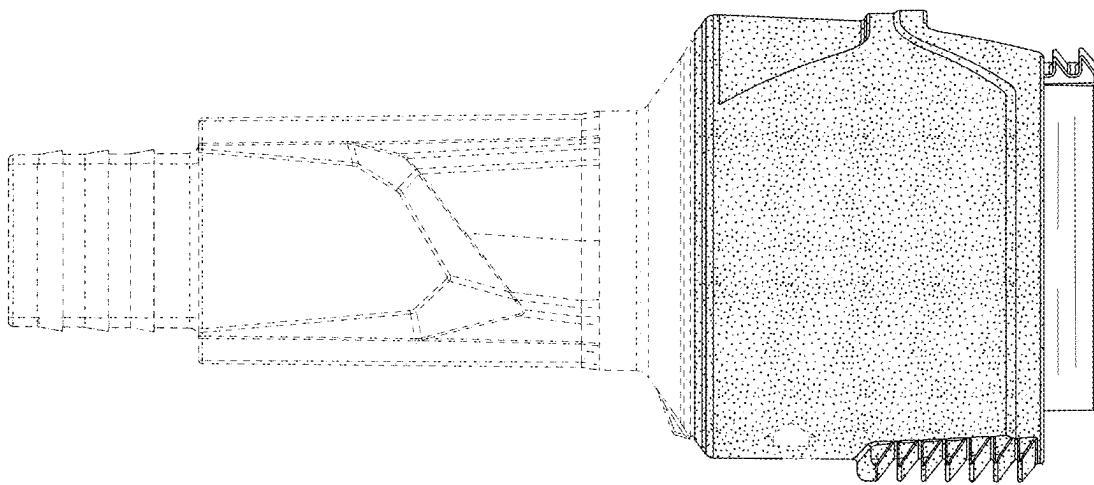


FIG. 12

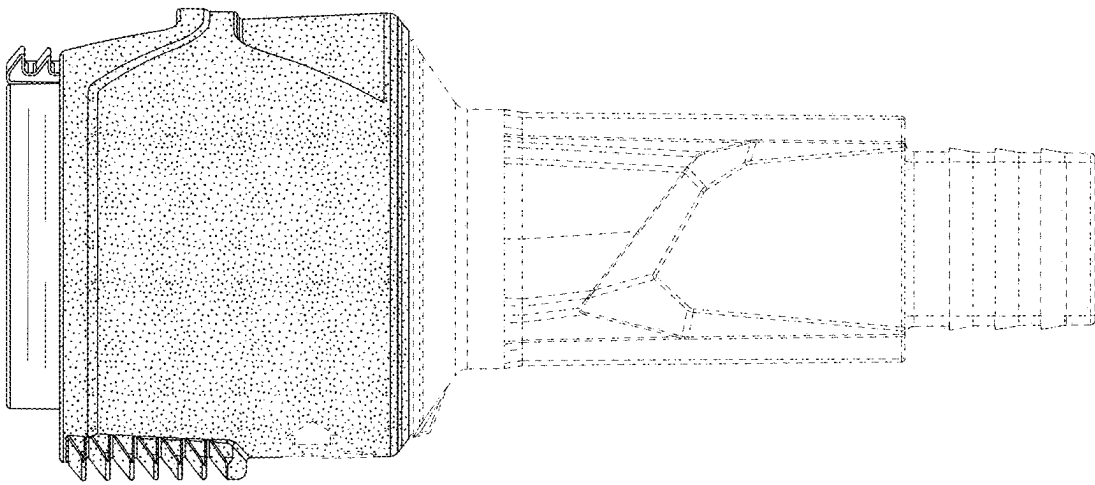


FIG. 13

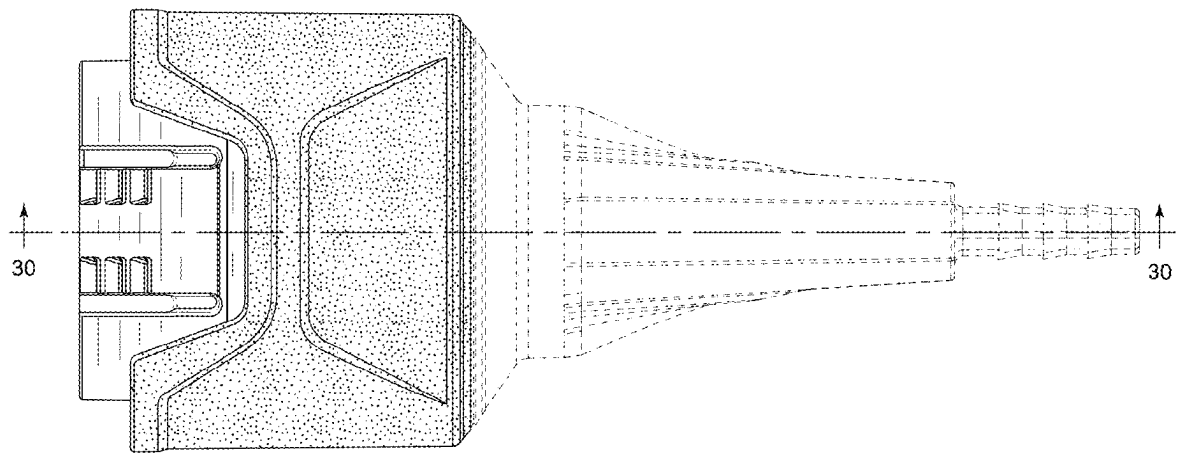


FIG. 14

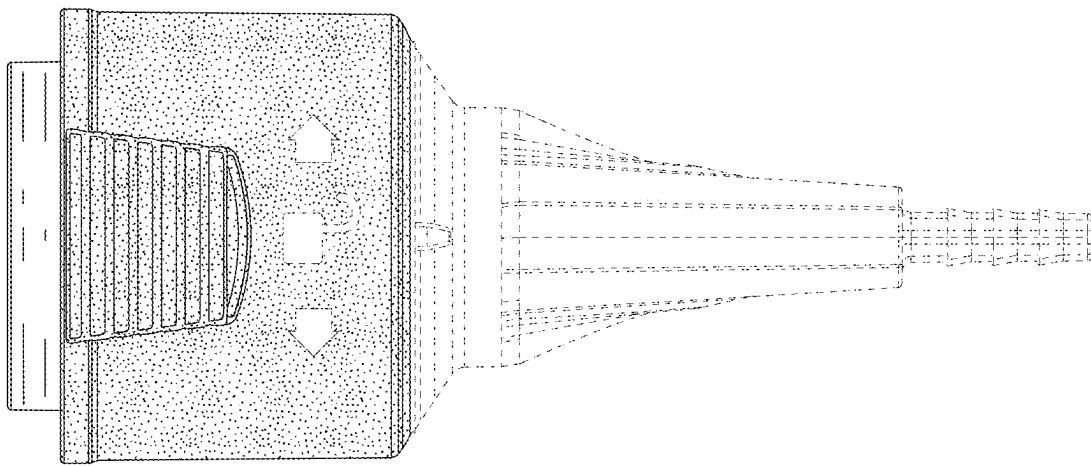


FIG. 15

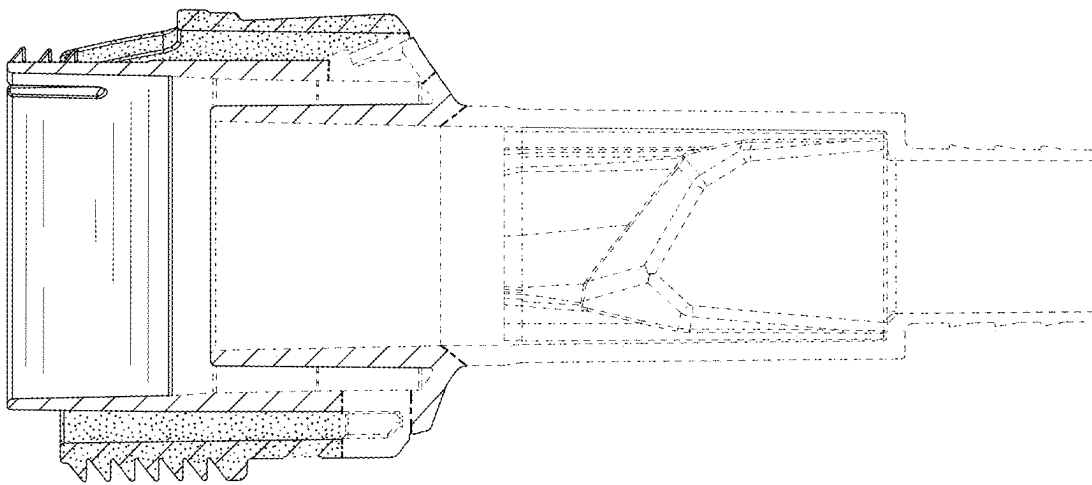


FIG. 16